Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building internal tools, data integrations, cloud applications, and applied AI workflows. Experienced in Python, TypeScript, React, Node.js, REST APIs, SQL, AWS, Azure, Docker, data pipelines, testing, automation, and production deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python APIs**, **Azure SQL** services, and React internal tools for 500+ users, translating stakeholder needs into maintainable workflows while reducing production latency from 1.2 seconds to 350 milliseconds.
- Automated 5,000+ monthly data workflows through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving observability, recovery, maintainability, operational resilience, and systematic production issue diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **React** interfaces and **Flask REST services** handling 10,000+ daily requests, implementing document workflows, authentication, progress states, and reusable components while resolving production defects and supporting 150+ users.
- Created multithreaded ingestion for 1,000+ files with validation, **AWS RDS** persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining responsive behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable **Python and PyTorch pipelines** processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, structured evaluation, performance monitoring, validation checks, and systematic debugging.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating outputs across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed scalable **Python services and data integrations** using AWS Glue, S3, SQL, filtering, pagination, and caching, improving production response times 30% and data-processing efficiency 68% across applications.
- Maintained 20+ automated unit and integration suites with **Jenkins**, PyTest, and Mockito, accelerating troubleshooting, release verification, and reliable cloud deployments across Windows, Linux, and macOS production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular **React and Node.js** tools across eight microservices with authentication, role-based access, responsive interfaces, and real-time updates, processing 2 million+ Azure SQL records through documented integrations.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through **Jenkins and Azure**, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

Skills

Languages & Full Stack: Python, TypeScript, JavaScript, Java, SQL, HTML5, CSS3, React, Next.js, Node.js, Flask, FastAPI, responsive web applications
Data & Integration: REST APIs, web services, WebSockets, microservices, data ingestion, data pipelines, ETL, PostgreSQL, Azure SQL, MySQL, MongoDB, AWS RDS, schema design, query optimization
Cloud & Automation: AWS, Azure, S3, AWS Glue, Docker, Kubernetes, Jenkins, CI/CD, cloud services, workflow automation, authentication, role-based access, distributed workflows
AI & Engineering: PyTorch, LLMs, semantic search, vector search, Git, unit testing, integration testing, functional testing, debugging, monitoring, code review, requirements analysis, documentation, production support, Agile

Projects

Big-Brain | Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*
- Built secure document ingestion, semantic search, and collaborative analysis with **LLaMA-2, FAISS, and REST APIs**, reducing review time 60% while supporting low-latency workflows across 50,000+ documents.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an **AI-powered data workspace** supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, metadata exploration, and Docker-backed deployment across modular Java and TypeScript services.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a responsive full-stack platform supporting 50,000+ users and one million transactions, integrating **secure APIs**, real-time WebSockets, MongoDB persistence, reusable interfaces, Kubernetes deployment, and automated release workflows.